

are being developed to improve potable water availability, are becoming a part of the market and households.

Water sensing

Applications of electronics in water-level indicators, controllers, switches and probes are already in play. Devices such as electronic level controllers, level sensors, water-level alarms, float switches and other types of water-level controllers have become an integral part of every household, and are helping in the conservation of water by controlling water wastage.

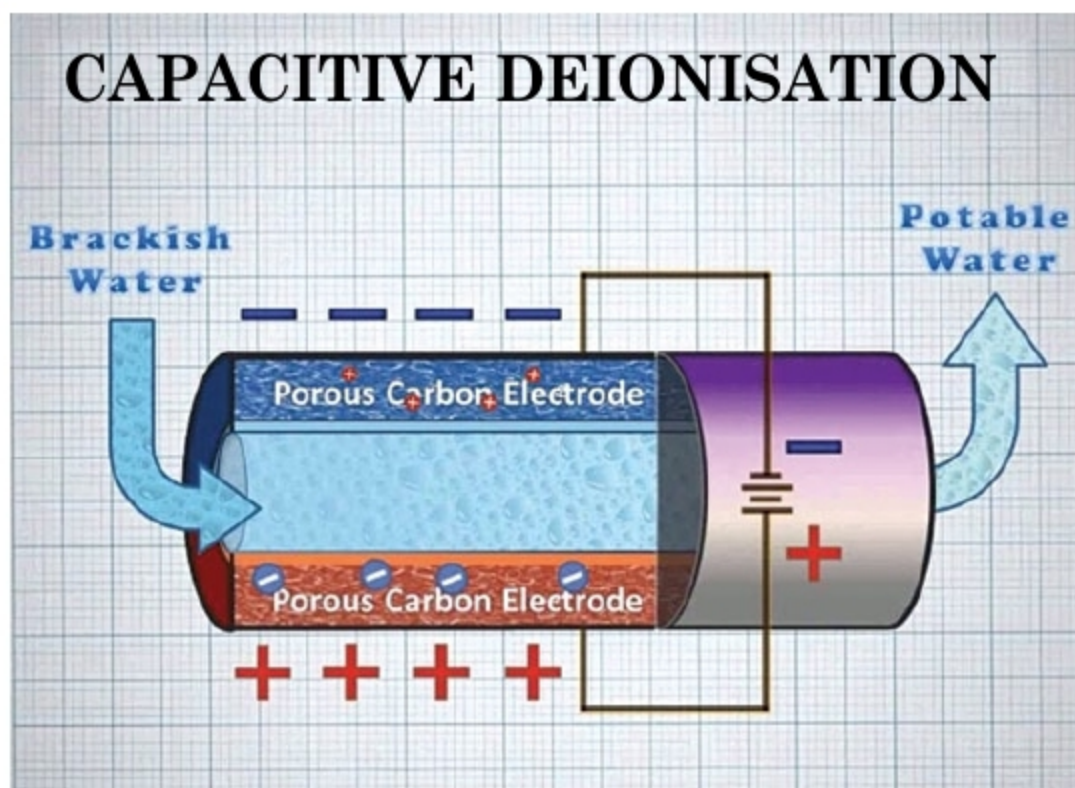
The electronic water-sensing system first senses the amount of water available in the water reservoir using the level detector and then adjusts state of the water pump in accordance with the water level. It achieves automation through sequential logic implemented using flip-flop circuits. A display and relay-based motor pump driving circuit are part of the integrated design. The pump automatically opens and stops for filling and emptying water.

Waterless washing

A waterless washing machine is being developed with the hope that it will stop water wastage. The washing machine's system is based on University of Leeds' Prof. Stephen Burkinshaw's research, which focuses on the structure of nylon polymer beads, because nylon is the best material for absorbing tiny particles.

He, along with his team of researchers, have come up with the concept of using nylon to remove stains from clothes. They have partnered with Xeros Ltd to produce waterless washing machines for commercial use. Currently, this technology is available at a small scale, but efforts are being made to scale it up and create a household version soon.

Another concept has been introduced by designer Elie Ahovi, which



Electronic water purification technology (Credit: www.youtube.com)

uses the principle of magnetic levitation. The new device consists of a spherical drum that cleans clothes by scrubbing them using dry ice, instead of water. The drum is essentially a ring filled with batteries, which produces magnetic levitation inside the ring and levitates the super-conductive metal laundry basket inside.

A low temperature is required inside the ring, which is produced by using liquid nitrogen. The entire process is controlled using a touchscreen interface. Dry ice, introduced inside the ring at supersonic speed, hits dirty clothes, and carbon-dioxide generated interacts with dirt and separates it from the clothes.

Dirt is then collected and filtered out through small pores, and carbon-dioxide is removed and re-frozen. This is how clothes are cleaned and dried.

Water purification

Electronic water purification enables users anywhere to convert readily-available but polluted non-salty water to potable water, at a fraction of the cost of bottled water. The system ensures that water that is used in business, industry and at home is safe to use.

The electronic water purification technology uses electromagnetism to rid water of impurities. The science behind the system is relatively straight-forward.

The electronic water purification method works by applying

DC power to carbon-based electrodes, which are layered into a casing. Application of this power causes the electrodes to be charged with opposing polarity. This opposing polarity attracts minerals present in the water, removing them from water. Minerals are absorbed on the surface of the electrodes, which causes all minerals and impurities to be removed, leaving behind nothing but pure water. When these contaminants

build up, they are simply dumped by reversing the polarity of the charge and then flushed away by a valve system.

Relatively cheap to purchase, install and maintain, the electronic water purifier is convenient and affordable to use. Whereas, other systems can be expensive and complicated. Because the electronic water purifier does not rely on high pressures to work, power requirements are low, making it efficient. It can even be used with a solar power system.

A high percentage of water is recovered, which makes the system more efficient than many other water purification technologies. The electronic water purifier requires no chemicals to operate, making it easy and safe to use. It easily removes more contaminants from water than other technologies, including total dissolved solids (TDSes), lead, nitrates, metals, perchlorate, sodium-chloride, sulphates, calcium and arsenic.

Since it can rid the water of contaminants that are potentially harmful, the electronic water purifier can be used to provide safe and clean drinking water for homes and businesses, as well as softening water for use in and around households for other tasks.

The electronic water purifier can also be used in a variety of commercial, industrial and large industry